FINAL PROJECT

FACULTY: SIT

PROGRAMME/DEPARTMENT: COMPUTER SCIENCE

OPTION:IIT	**LEVEL: III**
MODULE CODE:	**MODULE TITLE:** **Internet of Things**
DATE: 6-10 november, 2025	**SESSION: Day**
DURATION:5 days	**TOTAL MARKS:50 MARKS**
EXAMINER/LECTURER: **Joselyne Byukusenge**	**MODERATOR(S):**

INSTRUCTIONS TO THE CANDIDATE/STUDENT

1. Report Guidelines (20 Marks)(second assignment)

Each student must prepare a professional project report similar to a dissertation. The report should demonstrate your understanding of the project and your ability to organize technical information clearly.

Section	Description	Marks
Cover Page	Includes project title, student name(s), institution, supervisor, and submission date.	**2 Marks**
Abstract	Brief summary (5–10 lines) highlighting the purpose, methods, and results.	**2 Marks**
Table of Contents	Organized structure of the report with clear headings.	**1 Marks**
Introduction	Explains project background, problem statement, and objectives	**3 Marks**
Literature Review	Summarizes any related work and how your project builds upon it.	**3 Marks**
Methodology	Describes tools, technologies, and	**4 Marks**

	implementation steps used.	
Results and Discussion	Shows data, analysis, and interpretation of the project output(graphs, charts, screenshots with explanations)	**3 Marks**
Conclusion and references	Summarizes achievements, challenges, and future improvements you can do with all devices and more time. Again show source of your information	**2 Marks**

NB: Do Not Copy Others work

2. Prototyping and Practical Work (30 Marks)(Continuous assessment(CAT))

The practical assessment focuses on your ability to design, build, and demonstrate a functional IoT and embedded system prototype.

No.	Component	Description	**Marks**
1	Embedded Circuit Design	Circuit is neat, well connected, and all components function correctly.	**5 Marks**
2	Communication Protocol	Uses MQTT, HTTP, or CoAP to transmit data from device to cloud.	**10 Marks**
3	Custom API (No IoT Platforms)	Uses Python (Flask),PHP or Node-RED to send data to own server (not Blynk or ThingSpeak).	**10 Marks**
4	Device Control from IoT Platform	Implements remote control (e.g., LED, motor, or relay) via dashboard.	**5 Marks**

Submission and Presentation

All reports must be submitted by **Monday midnight 23:59**. Each student will present their project prototype and explain the data flow, circuit operation, and software implementation starting on **Tuesday. Prepare a single ppt presentation** to use in the presentation of your prototype.

Tip for Success

Show a complete understanding of both hardware and software components. Ensure your project runs smoothly from data collection to visualization. Creativity, clarity, and reliability will earn you the highest marks. **Good Luck!!!**